Learning to communicate across modalities: perceptual heterogeneity in multi-agent systems

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Emergent communication offers insight into how agents develop shared structured representations, yet most research assumes homogeneous modalities or aligned representational spaces, overlooking the perceptual heterogeneity of real-world settings. We study a heterogeneous multi-step binary communication game where agents differ in modality and lack perceptual grounding. Despite perceptual misalignment, multimodal systems converge to class-consistent messages grounded in perceptual input. Unimodal systems communicate more efficiently, using fewer bits and achieving lower classification entropy, while multimodal agents require greater information exchange and exhibit higher uncertainty. Bit perturbation experiments provide strong evidence that meaning is encoded in a distributional rather than compositional manner, as each bit's contribution depends on its surrounding pattern. Finally, interoperability analyses show that systems trained in different perceptual worlds fail to directly communicate, but limited fine-tuning enables successful cross-system communication. This work positions emergent communication as a framework for studying how agents adapt and transfer representations across heterogeneous modalities, opening new directions for both theory and experimentation.

1. Introduction

One research direction in language evolution seeks to explain how communication systems and structures can emerge from interaction among individuals without predefined linguistic conventions. Emergent Communication between artificial agents provides an experimental framework for investigating this process, allowing for observation into how agents develop and align communication systems through interaction under different environmental and cognitive constraints (Ren et al., 2020; Chaabouni et al., 2020; Lazaridou et al., 2018; Guo et al., 2019; Mihai & Hare, 2021; Spiegel et al., 2025). In most existing models, agents are assumed to share the same perceptual representations of their environment, meaning that they effectively ground their protocols (*e.g.*, symbols) in a common perceptual space (Harnad, 1990). This overlooks an important dimension of communication: interlocutors can inhabit different perceptual worlds and still establish successful communication (Hager & Krausa, 2019; Wang et al., 2021; Evtimova et al., 2018; Patel et al., 2021). In this work, we examine how perceptual misalignment between agents influences the structure and efficiency of emergent communication protocols. We show that such misalignment can lead to divergences in efficiency

and consistency, and that agents can adapt their protocols to communicate successfully across different perceptual spaces. By doing so, this work offers insight into how communication systems between interlocutors can emerge and persist even when the interlocutors differ in how they perceive and represent the world.

2. Experimental Setup

This work[1] builds on the multimodal, multi-step referential game introduced by Evtimova et al. (2018), extending it from an image-text to an audio-image setting. In this game, a *Sender* and *Receiver* observe different modalities of the same target object (*e.g.*, dog barking audio ↔ image of a dog) and must communicate to identify the correct referent among a set of candidates (Figure 1). At each timestep, the Sender emits a binary message, which the Receiver uses to decide whether to terminate the dialogue and guess the object, or to respond with a message. The exchange continues until the Receiver terminates the dialogue or the maximum step limit is reached. In this research, we study multimodal (audio→image) and unimodal (audio→audio) setups, assuming an audio-based Sender.

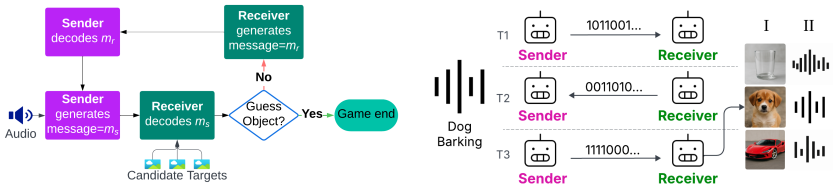


Figure 1. Illustration of the multi-step referential game, where a Sender encodes an audio input into a discrete message and a Receiver uses it to identify the correct target among a set of distractors—either images (I) in the multimodal case or audio (II) in the unimodal case—exchanging messages until a decision is made. Targets and distractor candidates are independently drawn at random from the corresponding datasets, with the possibility of sampling the same target in the unimodal case.

Agents. The Sender is a feed-forward neural network mapping an audio clip and the Receiver's latest message to a binary message (Figure 1, Supp.). The Receiver is a recurrent neural network that integrates its candidate inputs with the Sender's messages across timesteps, using this memory to decide whether to terminate and guess the target or to emit a binary message in response (Figure 2, Supp.).

Training and Data. The agents are jointly trained using RMSprop (Hinton, 2012) to minimise a combined objective integrating classification loss, REINFORCE-based reinforcement loss (Williams, 1992), and entropy regularisation. We use both a simple synthetic dataset called "Shapes World" for controlled evaluation and a natural environmental dataset (detailed in Supp. A.3).

[1]The code is available at https://github.com/naomipitzer/heterogeneous-emergent-communication. Further details on experimental setup in Supp. A

3. Results

We find that perceptual heterogeneity exerts a strong influence on the properties of the emergent language, namely on the efficiency, consistency, and interoperability of the learned protocols.

3.1. *Perceptual Misalignment and Efficiency*

We first question whether perceptual heterogeneity, manifesting as perceptual misalignment, decreases communication efficiency. Here, efficiency is defined as the ability of the agents to compress internal representations into short, discrete messages *while maintaining task accuracy*. To quantify this, message length is varied and the highest test accuracy achieved for both unimodal and multimodal systems is reported. Additionally, we measure the Receiver's classification entropy to capture uncertainty in message decoding under increasing compression.

Both systems tolerate compression up to a threshold (Table 1). However, when the message length is gradually reduced from 50 to 5 symbols, the systems' behaviour diverges. Unimodal agents maintain accuracy with low uncertainty, while **multimodal agents experience both degraded accuracy and increased uncertainty**. This reflects a modality gap, as misaligned representations add noise to the communication channel that lowers information transmitted per symbol, increasing the capacity required for reliable communication. (Shannon, 1948; Sucholutsky & Griffiths, 2023).

Table 1. Effect of message length in the Multimodal (M) and Unimodal (U) systems on communication efficiency and uncertainty.

Length	**Accuracy (%)**		**Loss**		**Rec Entropy**		**Sen Entropy**	
	M	U	M	U	M	U	M	U
1	34.00	36.00	1.82	1.81	1.19	1.03	16.95	14.98
5	70.00	87.00	0.89	0.22	1.06	0.53	16.70	15.18
10	81.00	82.00	0.64	0.47	0.68	0.55	20.75	20.77
30	83.00	83.00	0.62	0.34	0.66	0.45	21.37	21.38
50	85.00	87.00	0.53	0.36	0.67	0.48	21.48	21.62

3.2. *Class Consistency of Sender-Receiver Protocols*

A central property of a stable communication system is the consistent association between signals and meanings between interlocutors (Lewis, 1969; Steels, 2003; Lazaridou et al., 2017). Therefore, assessing message consistency across instances of a class allows us to test whether Senders and Receivers rely on shared representations. Consistency within a class is measured as the average pairwise cosine similarity among messages belonging to that class.

First, class-level consistency is measured separately for the Sender and Receiver to assess within-agent reliability. Sender messages exhibit strong class consistency, indicating convergence toward deterministic encodings (Figure 4, Supp.).

Receiver messages, in contrast, show weaker internal consistency. This difference is expected: Receiver outputs depend not only on the target but also on the distractor set and class confidence, which vary across trials. Additionally, Sender consistency increases as message length decreases, suggesting that tighter capacity constraints encourage more deterministic protocols. No such correlation was exhibited between Sender and Receiver messages, indicating that each agent develops its own protocol rather than relying on a common set of signals.

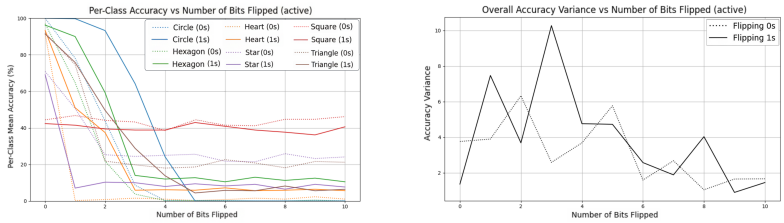
3.3. *Grounding and Semantics*

Having established that Sender and Receiver messages do not strongly correlate, we next examine how meaning is encoded within the Sender's messages and whether these encodings are grounded in perceptual input. All analyses are conducted on messages of fixed length 10. We use two approaches: perturbation experiments on message bits and message cluster analysis under varying perceptual features.

Bit Perturbation. We run a perturbation study in which bits that are almost always fixed (always 0 or always 1) and bits that vary are each flipped in turn to gauge their impact on classification performance (further details in Supp. B.3). These experiments reveal two key findings. Firstly, perturbing variable bits has minimal impact on accuracy in both unimodal and multimodal systems, while **flipping constant bits leads to sharp drops in accuracy**, indicating that they carry the most class-discerning information. However, perturbing variable bits in the unimodal case has a notably stronger impact on task accuracy than in the multimodal case, suggesting that these bits may encode the sender's fine-grained perceptual information that the multimodal Receiver, operating on a different modality, is less sensitive to (Figure 6, Supp.).

In addition, certain constant bits carry disproportionate information, as reflected by sharp accuracy declines and increased variance when they are perturbed (Figure 2). However, these bits do not appear to encode information compositionally (*i.e.*, their individual contributions cannot be combined to form interpretable meanings). For example, in the Heart class's case, flipping the final bit in the message from 0 to 1 reduced accuracy to zero, yet the same bit was consistently inactive across other classes. This indicates that the bit cannot be assigned a fixed class-specific meaning. Instead, its **informational role is determined by the surrounding bit pattern**.

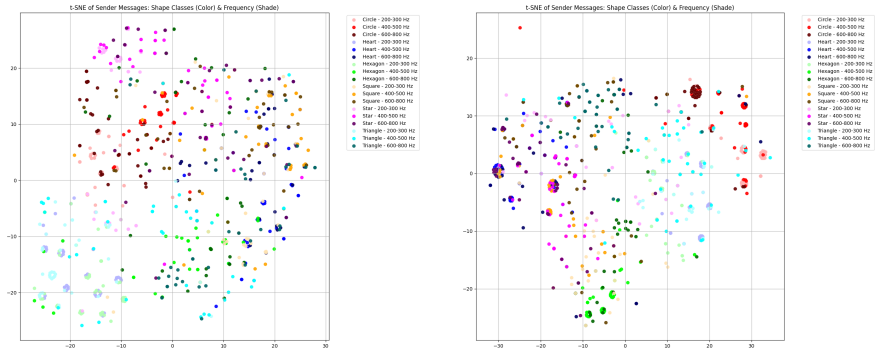
Message Cluster Analysis. To further investigate perceptual grounding, controlled subsets of the synthetic audio dataset are constructed in which either audio frequency or amplitude varies while the other feature is held constant. We visualise the resulting message representations using t-SNE (Maaten & Hinton, 2008). At the individual class level, frequency variation produces visible clusters: lower-frequency inputs form compact and separable groups (Figure 7, Supp.). Across



(a) Accuracy comparison (b) Variance comparison

Figure 2. Per-class accuracy and variance when flipping increasing numbers of constant bits. Accuracy drops more steeply when flipping 0 →1, and a variance spike around the third flipped 1-bit suggests certain active bits carry critical information.

the entire dataset, low-frequency inputs largely cluster in a shared region of the embedding space, indicating that frequency is encoded rather than randomly distributed (Figure 3). These effects are more pronounced in the unimodal system, yet remain visible in the multimodal system. These findings are notable, as they demonstrate that even when the Receiver perceives through a different modality, the Sender's messages continue to reflect perceptual structure grounded in its modality. In other words, the Sender's messages in multimodal communication **remain grounded in the Sender's perceptual space rather than becoming detached from it**.



(a) Unimodal (audio–audio) (b) Multimodal (audio–image)

Figure 3. t-SNE plots of **Sender message embeddings** for frequency variation. (a) In the unimodal system, messages cluster tightly by frequency. (b) In the multimodal system, clustering is weaker but still present, indicating grounding in perceptual frequency.

3.4. *Interoperability Across Modalities*

We next ask whether agents trained in different contexts can communicate effectively with one another. Specifically, we pair an audio-trained multimodal Sender

(originally trained with an image Receiver) with a unimodal audio Receiver. Without fine-tuning, accuracy falls to the level of random guessing (Table 2), providing strong evidence that the emergent protocols are not directly interoperable across systems. We then apply various levels of fine-tuning to the paired system. Beyond 2 epochs, accuracy increases substantially, demonstrating that minimal adaptation suffices to align the Sender's protocol with the unimodal Receiver. Extending fine-tuning further produces a trade-off, as accuracy on the newly paired system continues to improve, but at the cost of slightly degrading performance with the Sender's original partner. At around 15 epochs of fine-tuning, both systems achieve high task accuracy simultaneously (Figure 8, Supp.). Further details can be found in Supp. B.4.

Table 2. Comparison of classification accuracy between an audio-based Receiver system and image-based Receiver system across different training durations. With minimal fine–tuning, the sender is able to establish communication protocols with different Receivers.

Number of Epochs	Audio Receiver Accuracy (%)	Image Receiver Accuracy (%)
0	9.2	82.5
2	53.3	82.0
15	75.4	77.7
20	79.6	76.3
100	84.2	71.2

4. Conclusions and Future Directions

This study examined how perceptual input heterogeneity shapes the emergence, efficiency, and interoperability of communication protocols in artificial agents. The results show that when perceptual modalities are misaligned, the agents require longer messages and exhibit higher decoding uncertainty. Bit-level analyses further revealed that meaning is not localised to individual bits but distributed across the message pattern. Despite the input modality mismatch, the Sender's message embeddings remained grounded in its own perception, showing that communication can preserve perceptual structure even under representational divergence. Finally, we found that Senders can adapt their messages to communicate with Receivers experiencing differing perceptual realities. These findings highlight the asymmetries between unimodal and multimodal systems. Beyond the artificial settings, this work offers insight into how meaning and shared understanding can arise across divergent perceptual worlds, with implications for robotics, epistemology, and linguistic theories of embodiment and perceptual grounding. Future work should extend these analyses to alternative architectures, embodied agent systems (*e.g.*, robots), and different embedding-generation strategies to develop a broader account of heterogeneous emergent communication.

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